# DIAGNOSIS OF AUTISM SPECTRUM DISORDER USING MACHINE LEARNING TECHNIQUES

**A PROJECT REPORT**

***Submitted by***

**R. Savitha (RRN: 221421601049)**

**Under the guidance of**

**Dr. R. Sabin Begum**

Assistant professor

Department of Computer Applications

B.S. Abdur Rahman Crescent Institute of Science and Technology

# ABSTRACT

Autism Spectrum Disorder is a neurodevelopmental condition characterized by challenges in social interaction, communication, and behavior. Early and accurate diagnosis is crucial for effective intervention and support. In recent years, machine learning techniques have shows promising potential in the medical field, particularly in assisting with the diagnosis of complex disorders such as Autism Spectrum Disorder. This project aims to explore and evaluate the performance of five popular machine learning algorithms Support Vector Machine, Decision Tree, Random Forest, Naive Bayes, and Logistic Regression for the diagnosis of Autism Spectrum Disorder.

A dataset containing behavioral and demographic features relevant to Autism Spectrum Disorder screening is preprocessed and used to train and test each algorithm. Performance is measured using standard evaluation metrics including accuracy, precision, recall, and F1-score. The goal is to identify which algorithm provides the most reliable and interpretable results for Autism spectrum Disorder prediction. The findings of this study are intended to contribute toward developing supportive tools for clinicians and researchers working in the field of developmental disorders. Overall, the project demonstrates how machine learning can assist in improving diagnostic accuracy and supporting early intervention strategies for individuals with Autism Spectrum Disorder.

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## LIST OF ABBREVIATIONS

| ACRONYM | ABBREVIATION |
|---|---|
| ASD | Autism Spectrum Disorder |
| ML | Machine Learning |
| AI | Artificial Intelligence |
| SVM | Support Vector Machine |
| DT | Decision Tree |
| RF | Random Forest |
| NB | Naïve Bayes |
| LR | Logistic Regression |
| EDA | Exploratory Data Analysis |
| CSV | Comma-Separated Values |
| GUI | Graphical User Interface |
| API | Application Programming Interface |
| AUC | Area Under the Curve |
| ROC | Receiver Operating Characteristic |
| SHAP | SHapley Additive exPlanations |
| FL | Federated Learning |
| SMOTE | Synthetic Minority Over-sampling Technique |
| UCI | University of California, Irvine (Dataset Repository) |
| PCA | Principal Component Analysis |
| ANN | Artificial Neural Network |
| KNN | K-Nearest Neighbors |
| JSON | JavaScript Object Notation |

# CHAPTER 1

# INTRODUCTION

## 1.1 GENERAL

Autism Spectrum Disorder (ASD) is a lifelong neurodevelopmental condition that affects how individuals communicate, interact socially, and respond to their environment. The word "spectrum" refers to the wide range of symptoms, skills, and levels of disability that individuals with autism can experience. While some people with ASD may require significant support in their daily lives, others may function independently and possess exceptional skills in certain areas such as mathematics, music, or memory. Common signs of ASD include difficulties in social interaction, challenges with verbal and non-verbal communication, repetitive behaviors, and restricted interests. Individuals may avoid eye contact, have trouble understanding facial expressions or tone of voice, and may prefer routines or become distressed with changes in their environment. In addition, many people with ASD have unusual sensory responses, such as being highly sensitive to light, sound, or touch.

The exact causes of ASD are not yet fully understood, but research suggests a combination of genetic and environmental influences. Risk factors may include having a sibling with ASD, older parental age, genetic conditions such as Fragile X syndrome, and certain complications during pregnancy or birth. ASD is typically diagnosed in early childhood, often before the age of three, through developmental screenings and comprehensive evaluations by medical and psychological professionals. Standardized tools like the Autism Diagnostic Observation Schedule (ADOS) and clinical interviews are often used for diagnosis. According to recent statistics from the Centers for Disease Control and Prevention (CDC), approximately 1 in 36 children in the United States is diagnosed with ASD, with boys being more frequently affected than girls.

Although there is no cure for autism, early diagnosis and tailored intervention programs can significantly improve outcomes. Treatment plans often include a combination of behavioral therapies (such as Applied Behavior Analysis), speech and language therapy, occupational therapy, and educational support. These therapies focus on developing communication, social, and functional skills. With early support and consistent intervention, many individuals with ASD can lead productive and fulfilling lives. Increasing public awareness, reducing stigma, and improving access to services are essential in supporting individuals on the autism spectrum and their families. As research continues to advance, there is growing hope for more effective diagnostic tools and therapeutic strategies that can further enhance the lives of those affected by ASD.

## 1.2 EXISTING SYSTEM

Current diagnostic methods for Autism Spectrum Disorder (ASD) primarily rely on standardized tools such as the Autism Diagnostic Interview (ADI) and the Autism Diagnostic Observation Schedule (ADOS). While these methods are thorough and widely accepted, they come with several limitations. The diagnostic process is often time-consuming, sometimes taking over three years, which can delay critical early interventions. Additionally, these assessments require significant time and effort from trained professionals, making them resource-intensive and costly. In many regions, especially those with limited medical infrastructure, access to trained specialists is scarce, further hindering timely diagnosis. The manual and subjective nature of the diagnostic procedures also limits scalability and efficiency. Although machine learning (ML) has shows promise in healthcare applications, its use in ASD diagnosis is still emerging. Existing research is limited and often lacks comprehensive comparisons of different ML algorithms in real-world clinical settings, highlighting the need for more practical and scalable solutions.

### 1.2.1 Literature Survey

An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder (2024). Rownak Ara Rasul, Promy Saha, Diponkor Bala, S.M. Rakib Ul Karim, Md. Ibrahim Abdullah, Bishwajit Saha. This study explores the application of various machine learning models for the early diagnosis of Autism Spectrum Disorder (ASD) using children, adult, and combined datasets. Eight classification algorithms (including SVM, Logistic Regression, Random Forest, ANN, and others) are assessed using metrics such as accuracy, precision, recall, AUC, and log loss. The study also evaluates clustering techniques like k-means and spectral clustering to examine model performance without labeled data. Feature selection through chi-square testing and advanced preprocessing steps (e.g., missing value imputation, one-hot encoding, and feature scaling) enhance model accuracy. Results indicate that SVM and Logistic Regression achieved 100% accuracy for the children dataset, while Logistic Regression scored highest for the adult dataset (97.14%) and ANN performed best on the combined dataset (94.28%). A graphical user interface (GUI) using Streamlit was developed to support clinical use of the trained models.

Detection of Autism Spectrum Disorder in Children Using Machine Learning Techniques (2021). Kaushik Vakadkar, Diya Purkayastha, Deepa Krishnan. This study applies five machine learning algorithms—Logistic Regression, Naive Bayes, Support Vector Machine, K-Nearest Neighbors, and Random Forest Classifier—to detect Autism Spectrum Disorder (ASD) in children using the Q-CHAT-10 screening dataset. The data underwent preprocessing including label and one-hot encoding, followed by model training and evaluation. Logistic Regression achieved the highest accuracy at 97.15%, followed by Naive Bayes (94.79%), Support Vector Machine (93.84%), K-Nearest Neighbors (90.52%), and Random Forest (81.52%). The models were assessed using confusion matrices and F1 scores, with Logistic Regression identified as the most effective for the given dataset. The research highlights the potential of machine

learning in improving early ASD diagnosis and reducing reliance on lengthy clinical assessments

Machine Learning-Based Classification of Autism Spectrum Disorder across Age Groups (2024). Resmi Karinattu Reghunathan, Poornima Nanjagoundan Palayam Venkidusamy, Raju Gopalakrishna Kurup, Bindu George, Neetha Thomas.This study focuses on the application of machine learning algorithms to classify Autism Spectrum Disorder (ASD) across different age groups—children, adolescents, and adults—using datasets from the UCI repository. The authors used Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) models, with feature selection performed using the Cuckoo Search Algorithm. The study found that Logistic Regression outperformed other models in all age groups with accuracies of 97.01% (adult), 96.23% (child), and 89.42% (adolescent). KNN and SVM also showed strong performance but were slightly less accurate. The results support the potential of machine learning in aiding early ASD detection and diagnosis, especially when optimized through intelligent feature selection.

Prediction and Analysis of Autism Spectrum Disorder Using Machine Learning Techniques (2023). Muhammad Shuaib Qureshi, Muhammad Bilal Qureshi, Junaid Asghar, Fatima Alam, Ayman Aljarbouh. This study, which has since been retracted due to concerns about the peer-review process, focused on predicting Autism Spectrum Disorder (ASD) using machine learning algorithms including Random Forest (RF), Naive Bayes (NB), Support Vector Machine (SVM), and Multilayer Perceptron (MLP). The ABIDE dataset was used, and the data was preprocessed through techniques like SMOTE, outlier removal, and sequential forward feature selection. Among the models tested on the balanced dataset, Random Forest achieved the highest accuracy at 89.23%, followed by Naive Bayes (85.43%), MLP (81.84%), and SVM (80.43%). Despite the study's retraction, it originally aimed to offer a centralized framework for ASD prediction

and featured comparative analysis of performance metrics including precision and recall.

A Machine Learning Approach to Predict Autism Spectrum Disorder (ASD) for Both Children and Adults Using Feature Optimization (2023). Khandaker Mohammad Mohi Uddin, Hasibur Rahman, Mahadi Hasan, Fatema Akter, Suman Chandra Das. This paper presents a machine learning-based framework for predicting Autism Spectrum Disorder (ASD) in both children and adults using optimized features and publicly available datasets. The study applies six machine learning models—Random Forest (RF), Gaussian Naive Bayes (GNB), Bernoulli Naive Bayes (BNB), Multinomial Naive Bayes (MNB), Support Vector Machine (SVM), and Quadratic Discriminant Analysis (QDA)—on preprocessed and optimized datasets. Feature selection and missing value handling were used to improve performance. Results show that Random Forest achieved the highest accuracy (100%) on both child and adult datasets. GNB and BNB models also performed well, reaching up to 93% accuracy for children and 96% for adults. The findings support the use of Random Forest as a highly effective model for early-stage ASD prediction, surpassing other traditional classifiers.

Enhancing Early Detection of Autistic Spectrum Disorder in Children Using Machine Learning Approaches (2024). Rashid Ayub, Syed Immamul Ansarullah, Faizan Samdani, Ali Akhtar, Mohd Afzal, Shaik Moinuddin Ahmed. This study presents a machine learning-based diagnostic framework for early detection of Autism Spectrum Disorder (ASD) in children using a dataset from the UCI Machine Learning Repository. After preprocessing and chi-square feature selection, four classifiers—Random Forest, Logistic Regression, Gradient Boosting, and Extra Trees—were evaluated. The Random Forest model achieved 97% accuracy, while Logistic Regression reached 96%, and both Gradient Boosting and Extra Trees classifiers achieved 98% accuracy. All models demonstrated strong precision, recall, and F1-scores (≥96%) and AUROC values up to 1.00, indicating perfect classification performance in some cases. The study

concludes that ensemble models like Gradient Boosting and Extra Trees offer highly reliable ASD diagnostic support, and further recommends testing on larger datasets to validate scalability and clinical application.

Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning (2023). Muhammad Shoaib Farooq, Rabia Tehseen, Maidah Sabir, Zabihullah Atal. This study presents a novel federated learning (FL)-based approach for diagnosing Autism Spectrum Disorder (ASD) across children and adult populations using four publicly available datasets. FL allows models to be trained locally on each dataset to preserve data privacy while contributing to a central meta-classifier. Two machine learning models—Support Vector Machine (SVM) and Logistic Regression (LR)—were trained separately on child and adult datasets. The best-performing models were then combined using FL to form a global model. SVM achieved the highest accuracy of 99% on the children's dataset, while Logistic Regression reached 98%. For adults, SVM and LR achieved 81% and 78% accuracy, respectively. The study validates the models using performance metrics including precision, recall, F1-score, ROC curve analysis, and confusion matrices. The FL approach demonstrated high potential in securing sensitive healthcare data while enabling accurate, scalable ASD detection across age groups. The accuracy of the models indicates a significant advancement in the application of machine learning techniques for ASD diagnosis, highlighting the effectiveness of federated learning in maintaining data privacy while achieving high diagnostic performance

Autism Spectrum Disorder Research: Knowledge Mapping of Progress and Focus Between 2011 and 2022. Miaomiao Jiang, Tianlan Lu, Kang Yang, Xianjing Li, Liyang Zhao, Dai Zhang, Jun Li, Lifang Wang. This study employs bibliometric analysis to explore the evolution and current trends in autism spectrum disorder (ASD) research over the past decade. Utilizing data from the Web of Science Core Collection (WoSCC), the authors analyzed 57,108 publications from 2011 to 2022. The findings reveal a significant increase in

publications, with a 181.7% rise from 2,623 in 2011 to 7,390 in 2021. Key research clusters identified include "causative mechanisms," "clinical features," and "intervention features." The study highlights the growing focus on genetic variants associated with ASD, as well as emerging areas such as immune dysbiosis and gut microbiota. The bibliometric tools used include Bibliometrix, CiteSpace, and VOSviewer, which facilitated the visualization of research hotspots and trends. The results underscore the importance of interdisciplinary collaboration and suggest future research directions, particularly in understanding the microbe-gut-brain axis in relation to ASD.

Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning (2023). Muhammad Shoaib Farooq, Rabia Tehseen, Maidah Sabir, Zabihullah Atal. This study presents a novel federated learning (FL)-based approach for diagnosing Autism Spectrum Disorder (ASD) across children and adult populations using four publicly available datasets. FL allows models to be trained locally on each dataset to preserve data privacy while contributing to a central meta-classifier. Two machine learning models—Support Vector Machine (SVM) and Logistic Regression (LR)—were trained separately on child and adult datasets. The best-performing models were then combined using FL to form a global model. SVM achieved the highest accuracy of 99% on the children's dataset, while Logistic Regression reached 98%. For adults, SVM and LR achieved 81% and 78% accuracy, respectively. The study validates the models using performance metrics including precision, recall, F1-score, ROC curve analysis, and confusion matrices. The study demonstrates that the FL approach not only secures sensitive healthcare data but also enables accurate, scalable ASD detection across age groups.

Diagnosis of Autism Spectrum Disorder Based on Functional Brain Networks and Machine Learning (2023). Caroline L. Alves, Thaise G. L. de O. Toutain, Patricia de Carvalho Aguiar, Aruane M. Pineda, Kirstin Roster, Christiane Thielemann, Joel Augusto Moura Porto, Francisco A. Rodrigues. This study

presents a machine learning method for diagnosing Autism Spectrum Disorder (ASD) using functional brain imaging data from 500 subjects, including 242 with ASD. The approach utilizes the Bootstrap Analysis of Stable Clusters (BASC) map and achieves high accuracy, with an Area Under the Curve (AUC) near 1.0. Key findings include reduced connectivity in the left ventral posterior cingulate cortex among ASD patients. The methodology employs various classifiers, achieving a mean AUC of 0.99, with precision, recall, and F1-score all at 99%. The use of SHapley Additive Explanations (SHAP) values enhances the interpretability of the results, underscoring the potential of this approach in clinical settings.

### 1.2.2 Limitations of Existing System

**a) Delayed Diagnosis**

- Traditional diagnostic tools like ADI and ADOS often result in a diagnosis delay of up to three years, which can severely impact the effectiveness of early interventions.

**b) Resource-Intensive**

- These methods require significant human effort, involving trained clinicians and extended observation periods, making them laborious and expensive.

**c) Limited Accessibility**

- Access to these diagnostic tools is often restricted, especially in rural or underserved regions, due to a shortage of qualified professionals and infrastructure.

**d) Manual and Non-Automated**

- The diagnosis process heavily relies on manual assessment and interpretation, lacking automation, which limits scalability and consistency.

#### e) Inconsistent Results

- Due to the subjective nature of observations and assessments, diagnostic outcomes may vary across clinicians and settings

## 1.3 PROPOSED SYSTEM

The proposed system aims to enhance the early diagnosis of Autism Spectrum Disorder (ASD) by leveraging various machine learning algorithms. Unlike traditional diagnostic methods that are time-consuming and resource-intensive, this system automates the diagnostic process, making it faster, more accurate, and accessible. The approach involves the use of supervised machine learning models such as Support Vector Machine (SVM), Random Forest, Decision Tree, Logistic Regression, and Naive Bayes to classify individuals based on clinical and behavioral data. The system begins with data collection from publicly available ASD screening datasets, followed by preprocessing steps including handling of missing values, encoding of categorical variables, and feature normalization.

### 1.3.1 Advantages of Proposed System

#### a) Early and Accurate Diagnosis

- The use of machine learning enables early detection of ASD by identifying patterns in behavioral and clinical data with high accuracy, which is crucial for timely intervention.

#### b) Time-Efficient

- Unlike traditional diagnostic methods that may take years, this system provides quick results, significantly reducing the time required for diagnosis.

c) **Cost-Effective**

- By automating the diagnostic process, the system reduces dependency on extensive professional labor, lowering overall healthcare cost.

d) **Scalability**

- The system can be easily scaled to screen large populations, making it suitable for mass screening and public health initiatives.

e) **Automation and Consistency**

- Machine learning models eliminate subjective human bias and ensure consistent decision-making, improving diagnostic reliability.

## 1.4 ORGANIZATION OF THE CHAPTERS

The work done in various phases has been organized into chapters. Chapter 1 gives an introduction to Autism Spectrum Disorder (ASD) and explains why early diagnosis is important. Chapter 2 clearly defines the research problem and presents the methodology adopted for the project. Chapter 3 describes Development Process focuses on the foundational elements required to build the ASD diagnosis system. Chapter 4 describes Report Generation outlines the practical implementation and modular breakdown of the Autism Spectrum Disorder (ASD) diagnosis system. Chapter 5 presents the experimental results, performance evaluation of the models using various metrics, and a comparison between Algorithms, and suggests directions for future improvement.

# CHAPTER 2

# PROBLEM DEFINITION AND METHODOLOGY

## 2.1 PROBLEM DEFINITION

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that is traditionally diagnosed through time-consuming and resource-intensive behavioral assessments. The problem lies in the inefficiency and delayed diagnosis associated with current clinical methods. The objective is to leverage machine learning algorithms to develop a faster, more cost-effective, and accurate system for early ASD diagnosis by analyzing behavioral and demographic features, thereby improving timely intervention and patient outcomes.

## 2.2 METHODOLOGY

This project "Diagnosis of Autism Spectrum Disorder using Machine Learning Techniques," the methodology involves several key steps. First, a dataset containing relevant features about children, such as age, gender, and behavioral indicators, is collected. The data is then preprocessed, including handling missing values, feature selection, and normalization. Afterward, Five machine learning algorithms—SVM, Decision Tree, Random Forest, Naive Bayes, and Logistic Regression—are trained on the data, with hyper parameter tuning and cross-validation to optimize their performance. The models are evaluated using metrics like accuracy, precision, recall, and F1-score. The best-performing model is selected, and the final diagnosis system is deployed on a user-friendly interface, providing real-time ASD predictions.

### 2.2.1 Machine Learning

Machine learning is a branch of artificial intelligence that allows computers to learn from data and make decisions without explicit programming. It involves training algorithms on datasets to identify patterns and make predictions or classifications. There are three main types of machine learning: supervised

learning (where the algorithm learns from labeled data), unsupervised learning (where it finds patterns in unlabeled data), and reinforcement learning (where an agent learns through rewards and penalties). Machine learning models like SVM, Naïve Bayes, Logistic regression, Decision Trees, and Random Forests are used to solve various problems by predicting outcomes based on input data.

### 2.2.2 Support Vector Machine

Support Vector Machine (SVM) is a supervised machine learning algorithm primarily used for classification tasks, though it can also be used for regression. It works by finding the optimal hyperplane that best separates the data into different classes. The goal of SVM is to maximize the margin, or distance, between the closest data points (support vectors) from each class, ensuring the most robust separation between them. SVM can handle both linear and non-linear classification by using different types of kernels, such as the linear kernel (for linearly separable data) or the radial basis function (RBF) kernel (for non-linear data). SVM is effective in high-dimensional spaces and is widely used in applications like image recognition, text classification, and bioinformatics. This figure 2.1 shows how support vector machine works.

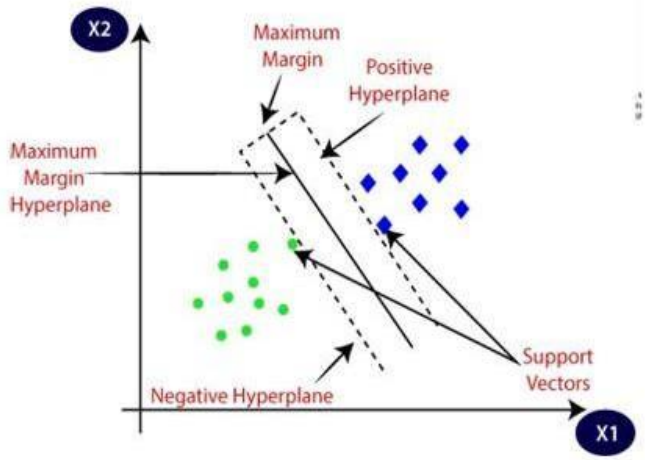


**Figure 2.1: Architecture diagram for Support Vector Machine**

### 2.2.3 Decision Tree

A decision tree is a popular machine learning algorithm used for both classification and regression tasks. It is a tree-like model that makes decisions based on the features of the input data. Here's a breakdown of how decision trees work, their advantages, disadvantages, and some key concepts. This figure 2.2 shows how decision tree works.

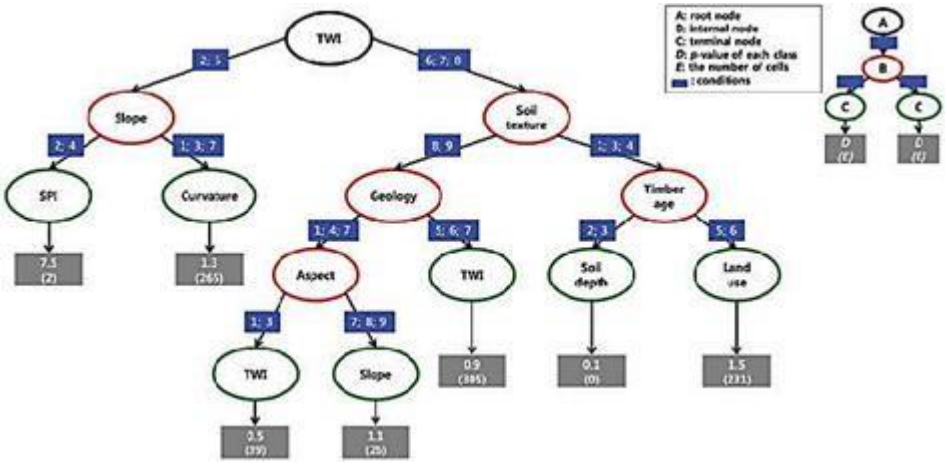


**Figure 2.2: Architecture diagram for Decision tree**

### 2.2.4 Naive Bayes

Naive Bayes is a family of probabilistic algorithms based on Bayes' Theorem, used primarily for classification tasks. It is particularly effective for large datasets and is known for its simplicity and efficiency. The "naive" aspect of the algorithm comes from the assumption that the features are independent given the class label, which is often not the case in real-world data. Despite this assumption, Naive Bayes can perform surprisingly well in practice. Naive Bayes classifier assumes that the effect of a particular feature in a class is independent of other features. For example, a loan applicant is desirable or not depending on his/her income, previous loan and transaction history, age, and location. This figure 2.3 shows how Naïve Bayes works.

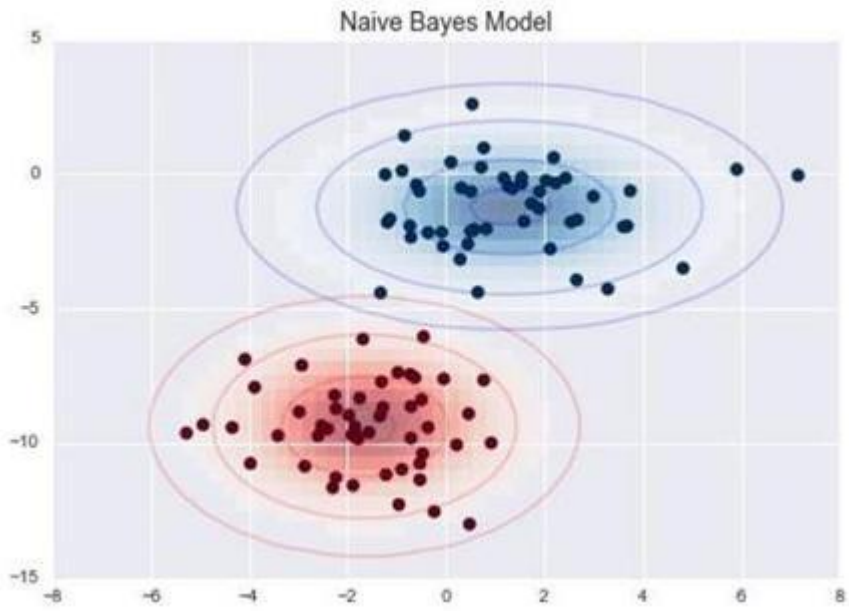


**Figure 2.3: Architecture diagram for Naïve Bayes**

### 2.2.5 Random Forest

A random forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and control overfitting. It is widely used for both classification and regression tasks, leveraging the diversity of trees to enhance model performance. This figure 2.4 shows how Random forest works.

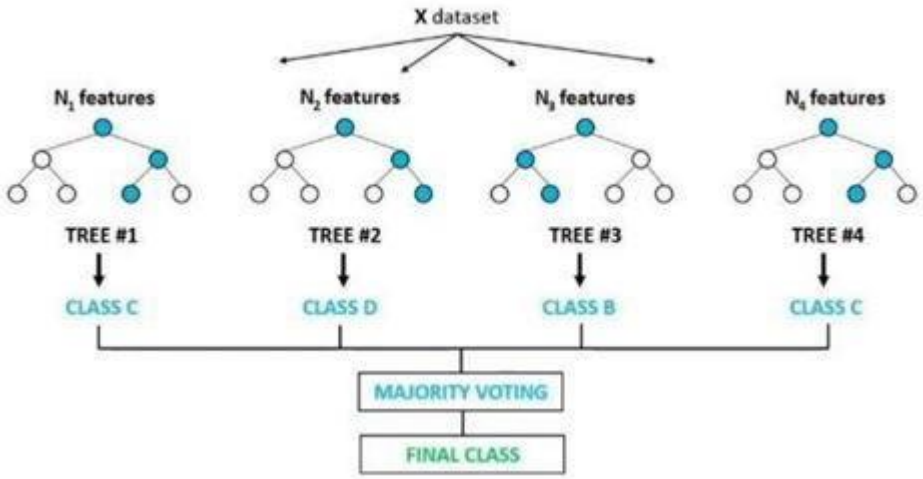


**Figure 2.4: Architecture diagram for Random Forest**

### 2.2.6 Logistic Regression

Logistic regression is a statistical method used for binary classification problems in machine learning. It models the relationship between a dependent binary variable and one or more independent variables by estimating probabilities using a logistic function. Despite its name, logistic regression is used for classification tasks rather than regression tasks. This figure 2.5 shows how Logistic Regression works.

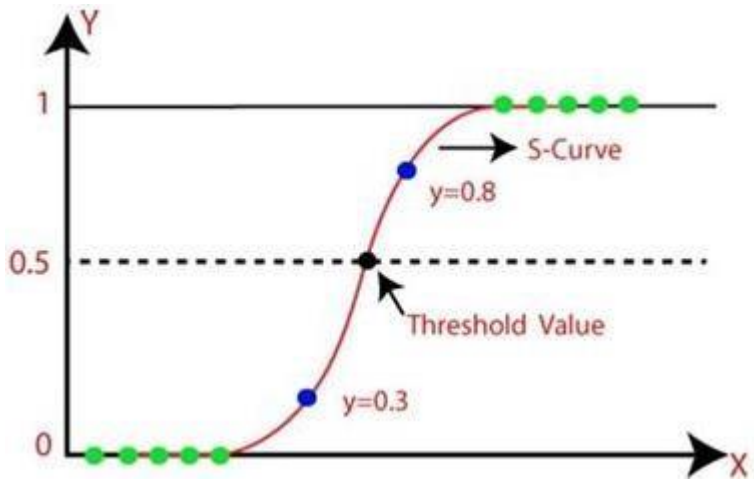


**Figure 2.5: Architecture diagram for Logistic Regression**

## 2.3 PYTHON MODULES

In Python, a module is a file containing Python code that defines functions, classes, and variables. Modules allow you to organize and reuse code across different programs**.**

### 2.3.1 Pandas

Pandas is a powerful and easy-to-use Python library for data analysis and manipulation. It provides two primary data structures—Series (1D) and DataFrame (2D)—which allow users to handle structured data efficiently. With pandas, you can load data from various sources like CSV or Excel files, clean and process missing values, filter or group data, and perform statistical operations with ease. It is widely used in data science and machine learning projects for preparing datasets before applying models.

### 2.3.2 Numpy

NumPy (Numerical Python) is a fundamental Python library used for scientific computing and working with large, multi-dimensional arrays and matrices. It provides a powerful array object called ndarray, along with a wide range of mathematical functions to operate on these arrays efficiently. NumPy is known for its speed and performance, especially in handling numerical data. It is often used in data analysis, machine learning, and scientific applications where fast numerical computations are essential. Although not directly used in your code, it is typically imported as a standard tool in data science projects for tasks like mathematical operations, array reshaping, and integration with other libraries like pandas and scikit-learn.

### 2.3.3 Scikit-Learn

Scikit-learn is a powerful and easy-to-use Python library for machine learning. It provides simple tools to build, train, and evaluate models for tasks like classification, regression, and clustering. Scikit-learn also includes functions for splitting data, preprocessing (like scaling and encoding), and measuring performance using accuracy, confusion matrices, and more. It's widely used in data science projects because of its clean syntax and strong integration with libraries like pandas and NumPy

**a) Model Selection**

The model selection module in Scikit-learn provides tools to split datasets, evaluate models, and fine-tune their performance. One of its most commonly used functions is train testsplit(), which divides data into training and testing sets. It also supports cross-validation techniques to check how well a model generalizes to unseen data. Additionally, tools like GridSearchCV help in finding the best parameters for a model by testing multiple combinations. Overall, model selection is essential for building reliable and optimized machine learning model.

**b) Pre processing**

The preprocessing module in Scikit-learn provides tools to prepare and transform data before feeding it into a machine learning model. It includes methods for scaling, normalizing, encoding, and handling missing values. For example, Standard Scaler is used to standardize features by removing the mean and scaling to unit variance, which helps models perform better. Label Encoder is used to convert categorical labels into numeric form. These preprocessing steps are crucial to ensure the data is in the right format and scale for machine learning algorithms to work effectively.

**c) Metrics Accuracy:**

Accuracy is the most intuitive performance measure and it is simply a ratio of correctly predicted observation to the total observations.

- **Precision:** Precision is the ratio of correctly predicted positive observations to the total predicted positive observations.
- **Recall:** Recall is the ratio of correctly predicted positive observations to the all observations in actual class.
- **F-Measure:** F-Measure Score is the weighted average of Precision and Recall. Therefore, this score takes both false positives and false negatives into account. It represents the performance measure of proposed system, from the pre-processed data, some feature vector methods and some classifiers has been processed, based on these methods the above table has been created.

# CHAPTER 3

# DEVELOPMENT PROCESS

Development process is one of the critical activities in software development life cycle. This is the phase where the requirements are analyzed. This includes both product requirements and resource requirements.

## 3.1. REQUIREMENT ANALYSIS

Requirements are a feature of a system or description of something that the system is capable of doing in order to fulfil the system's purpose. It provides the appropriate mechanism for understanding what the customer wants, analysing the needs assessing feasibility, negotiating a reasonable solution, specifying the solution unambiguously, validating the specification and managing the requirements as they are translated into an operational system.

### 3.1.1 PYTHON

Python is a dynamic, high level, free open source and interpreted programming language. It supports object-oriented programming as well as procedural oriented programming. In Python, we don't need to declare the type of variable because it is a dynamically typed language.

## 3.2 ANACONDA NAVIGATOR

Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda distribution that allows users to launch applications and manage anaconda packages, environments and channels without using command-line commands. Navigator can search for packages on Anaconda Cloud or in a local Anaconda Repository, install them in an environment, run the packages and update them. It is available for Windows, macOS and Linux.

### 3.2.1 JUPYTER NOTE BOOK

Jupyter Notebook (formerly IPython Notebooks) is a web-based interactive computational environment for creating Jupyter notebook documents. The

"notebook" term can colloquially make reference to many different entities, mainly the Jupyter web application, Jupyter Python web server, or Jupyter document format depending on context. A Jupyter Notebook document is a JSON document, following a versioned schema, containing an ordered list of input/output cells which can contain code, text (using Markdown), mathematics, plots and rich media, usually ending with the ".ipynb" extension. By default, Jupyter Notebook ships with the IPython kernel. As of the 2.3 release[11][12] (October 2014), there are currently 49 Jupyter compatible kernels for including Python, R, Julia and Haskell. many programming languages, The Notebook interface was added to IPython in the 0.12 release[14] (December 2011), renamed to Jupyter notebook in 2015 (IPython 4.0 – Jupyter 1.0). According to The Atlantic, Jupyter interest overtook the popularity of the Mathematica notebook interface in early 2018.

### 3.3 RESOURCE REQUIREMENTS:

The hardware and software requirements for this system were analyzed and the required configuration is given below.

#### 3.3.1 HARDWARE REQUIREMENTS:

Software engineers use the hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system as the starting point for the system design. It shows what the system does, and it should be implemented. The minimum Hardware Requirement are shows in figure 3.1.

**Table 3.1: Hardware requirements**

| | |
|---|---|
| Processor | Intel(R) Core(TM) i5 |
| Speed | 2.00 GHz |
| Memory | 4.00 GB |
| Display | 15” Monitor |
| Storage | 237 GB |
| Keyboard | 104 Keys |
| Graphics Card | CPU&GPU |

### 3.3.2 SOFTWARE REQUIREMENTS:

The software requirements are description of features and functionalities of the target system. Requirements convey the expectations of users from the software product. The requirements can be obvious or hidden, known or unknown, expected or unexpected from client's point of view. The minimum Software Requirements are shows in figure 3.2.

**Table 3.2: Software requirements**

| | |
|---|---|
| Operating System | Windows 10 |
| Tools | Anaconda Navigator<br>Jupyter Notebook |
| Language | Python |

## 3.4 DESIGN

Design is the process of planning and creating with intention, aiming to solve a problem or fulfill a specific purpose by defining the structure, components, and appearance of a system, object, or process.

### 3.4.1 System Design

System design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. System design in a Machine Learning (ML) project refers to the overall architecture and planning of how the different components of the ML system will work together to solve a specific problem efficiently, reliably, and at scale. It is a process of problem solving and planning for a software solution. There are different levels of designs. The figure 3.1 shows the system architecture diagram.

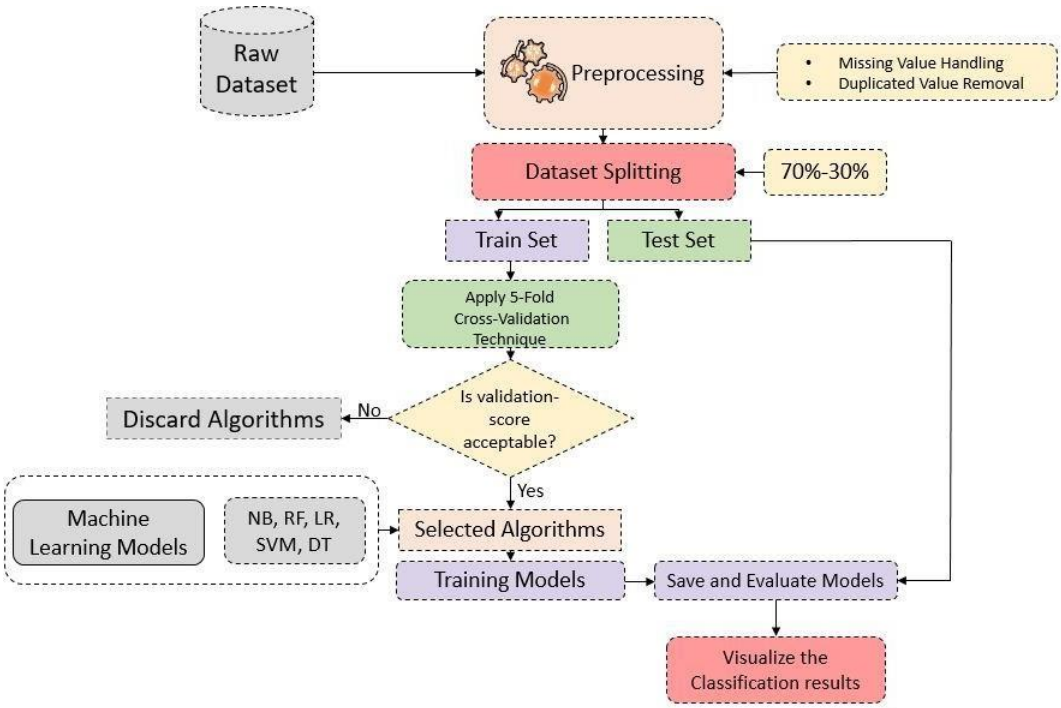


**Figure 3.1: System Architecture diagram**

### 3.4.2 FLOW DIAGRAM

The flow diagram 3.2 represents the process involved in diagnosis of autism spectrum disorder using Machine Learning techniques.

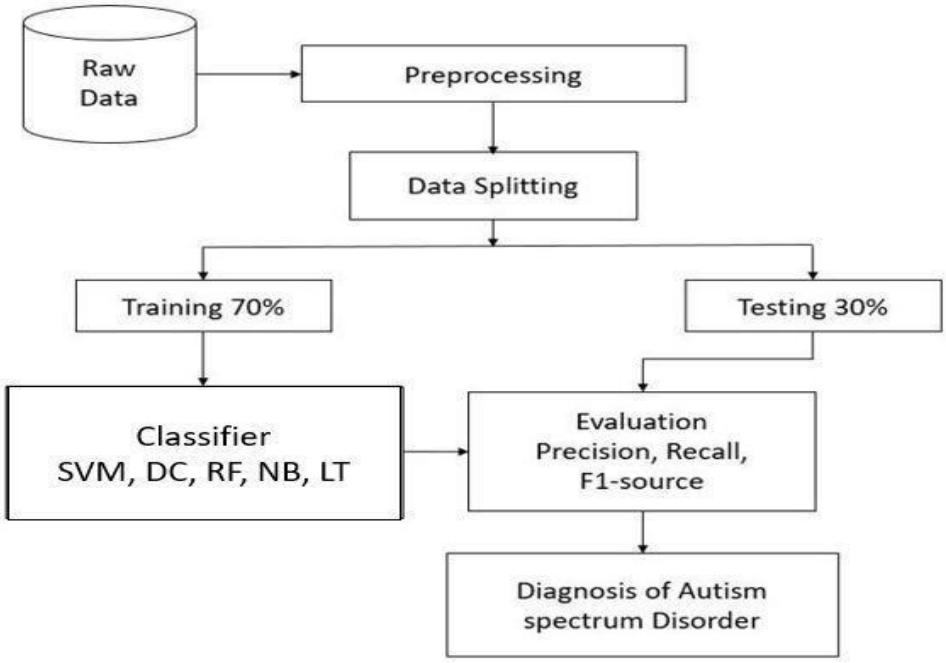


**Figure 3.2: Flow Diagram**

## 3.5 IMPLEMENTATION

Implementation includes all those activities that take place to convert from old system to the new. The new system may be totally new, replacing an existing system or it may be major modification to the system currently put into use. In this project, design is done by using the Python

### 3.5.1 Data Set

The dataset "ASD Screening Data for Children" contains 28 attributes, including categorical, binary, and continuous types. It consists of 1,985 instances in total, with 1,074 positive (Yes) and 911 negative (No) cases. This dataset is specifically designed for early detection of Autism Spectrum Disorder (ASD) in children. It provides a rich foundation for building and evaluating machine learning models for ASD diagnosis. Table 3.3 shows the preview of the dataset.

**Table 3.3: Preview of the Dataset**

| Sr. no. | Dataset name | Attribute type | Number of attributes | Number of instances | | |
|---|---|---|---|---|---|---|
| | | | | Yes | No | Total |
| 01 | ASD Screening data for Children | Categorical, binary and continuous | 28 | 1,074 | 911 | 1,985 |

### 3.5.2 Training Data and Test Data

Machine learning concept includes learning some information or we can say properties from dataset and testing properties of another data can be classified. Mostly we are splitting data in two categories as training set and testing set. Training set is the set of learning properties from given data. Testing set is to check properties of the data given for query which may include some different properties. Depending on Training set with different classifier we may get different results with different accuracy. The classifier of type deep learning which is having highest classification accuracy can be considered for further analysis. The below figure 3.3 shows the Data Training and Testing Steps.

```
[18]: # Define features and target variable
      X = df.drop(columns=['class'])
      y = df['class']
```

```
[19]: # Split dataset into training and testing sets
      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
[20]: # Normalize features
      scaler = StandardScaler()
```

```
[21]: X_train = scaler.fit_transform(X_train)
      X_test = scaler.transform(X_test)
```

**Figure 3.3: Training and Testing Data**

## 3.6 MODULES

The modular design defines the structure of all the modules in the system. This concept breaks the system into separate components to build parallelly and to concentrate on every small detail and requirement. And finally, all the modules are combined or connected together to work.

### 3.6.1 Data Preprocessing

Before training the models, the data is cleaned and prepared. This step is called data preprocessing. In your project, duplicate rows are first removed to avoid repeated data. Then, missing values are handled. After that, Label Encoding is applied to convert text labels (like "yes", "no") into numbers, so the machine learning models can understand them. This step is very important to make sure the data is clean, consistent, and ready for training. The below figure 3.4 shows the Data Preprocessing Steps.

```
# Load the dataset
df = pd.read_csv('data_csv.csv')
```

```
# Remove duplicate rows
df.drop_duplicates(inplace=True)
```

```
# Handle missing values (fill with mode for categorical and median for numerical)
for col in df.columns:
    if df[col].dtype == 'object':
        df[col].fillna(df[col].mode()[0], inplace=True)
    else:
        df[col].fillna(df[col].median(), inplace=True)
```

```python
# Encode categorical variables
label_encoders = {}
for col in df.select_dtypes(include=['object']).columns:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    label_encoders[col] = le
    print(f'Label Encoder mapping for {col}: {dict(zip(le.classes_, le.transform(le.classes_)))}')
```

```
Label Encoder mapping for Speech Delay/Language Disorder: {'No': 0, 'Yes': 1}
Label Encoder mapping for Learning disorder: {'No': 0, 'Yes': 1}
Label Encoder mapping for Genetic_Disorders: {'No': 0, 'Yes': 1}
Label Encoder mapping for Depression: {'No': 0, 'Yes': 1}
Label Encoder mapping for Global developmental delay/intellectual disability: {'No': 0, 'Yes': 1}
Label Encoder mapping for Social/Behavioural Issues: {'No': 0, 'Yes': 1}
Label Encoder mapping for Anxiety_disorder: {'No': 0, 'Yes': 1}
Label Encoder mapping for Sex: {'F': 0, 'M': 1}
Label Encoder mapping for Ethnicity: {'Asian': 0, 'Black': 1, 'Hispanic': 2, 'Latino': 3, 'Middle Eastern': 4, 'Mixed': 5, 'Native Indian': 6, 'Others':
7, 'PaciFica': 8, 'South Asian': 9, 'White European': 10, 'asian': 11, 'black': 12, 'middle eastern': 13, 'mixed': 14, 'south asian': 15}
Label Encoder mapping for Jaundice: {'No': 0, 'Yes': 1}
Label Encoder mapping for Family_mem_with_ASD: {'No': 0, 'Yes': 1}
Label Encoder mapping for Who_completed_the_test: {'Family Member': 0, 'Family member': 1, 'Health Care Professional': 2, 'Others': 3, 'School and NGO':
4, 'Self': 5}
Label Encoder mapping for class: {'No': 0, 'Yes': 1}
```

**Figure 3.4: Data Preprocessing**

### 3.6.2 Model Training and Evaluation

After preparing the dataset, the next step is to train the machine learning models. The data is first split into training and testing sets, so the model can learn from one part and be tested on the other. Five different algorithms are used for training: Support Vector Machine (SVM), Decision Tree, Random Forest, Logistic Regression, and Naive Bayes. Each model learns patterns from the training data.

### 3.6.3 Model Evaluation

After training the models, their performance is evaluated to check how well they can diagnose Autism Spectrum Disorder (ASD). This is done using evaluation metrics like accuracy score, confusion matrix, and classification report. The accuracy score tells how many predictions were correct. The confusion matrix shows the number of correct and incorrect predictions in detail. The classification report provides useful values like precision, recall, and F1-score, which help understand the strengths and weaknesses of each model. These evaluation results help compare the five algorithms—SVM, Decision Tree, Random Forest, Logistic Regression, and Naive Bayes—and identify the best one for ASD diagnosis. Figure 3.5 shows the model evaluation step.

```python
# Make predictions
y_pred = lr_model.predict(X_test)
```

```python
# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy:.2f}')
print(classification_report(y_test, y_pred))
```


```python
# Make predictions
y_pred = lr_model.predict(X_test)
```

```python
# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy:.2f}')
print(classification_report(y_test, y_pred))
```


**Figure 3.5: Model Evaluation**

### 3.6.4 Data Visualization

Data visualization helps in understanding the data and the performance of the models more effectively. In this project, matplotlib and seaborn libraries are used to create graphs and plots. A heatmap is used to display the confusion matrix, which gives a clear view of how well the model is classifying ASD and Non-ASD cases. Visual tools make it easier to compare results across different models. Scatter plots or bar charts are also used to show model accuracies and feature distributions, making the results more understandable and visually appealing. These visualizations play an important role in analyzing the overall performance of the five algorithms used in the project. Figure 3.6 shows the Data visualization step.

```python
# Plot confusion matrix
plt.figure(figsize=(6,4))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=label_encoders['class'].classes_, yticklabels=label_encoders['class'].classes_)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

**Figure 3.6: Data visualization**

# CHAPTER 4

# RESULT AND ANALYSIS

## 4.1 RESULTS

The results of this project demonstrate the effectiveness of machine learning (SVM, Decision Tree, Random Forest, Naïve bayes and Logistic regression) in detecting bacterial blight disease in cotton leaves. The evaluation metrics—accuracy, precision, recall, F1-score and Support —were computed to assess the performance of each model.

### 4.1.1. Support Vector Machine

SVM is useful in solving classification as well as regression problem statements. In the classification problem, two classes are separated or classified by a hyperplane. SVM creates two marginal lines along with a hyperplane having some distance so that they will be easily linearly separable for both the classification points. Along with the hyperplane, two parallel planes are also created passing to the nearest point of the two classes (support vectors) respectively. The distance between the planes is known as the marginal distance which acts as a cushion to divide a point into classes in a better way. The best hyperplane is selected which has the maximum marginal distance. The Accuracy, Precision, Recall, F1-Score and Support of SVM algorithm are shows in the Table 4.1 and the Confusion Matrix is shows in the Figure: 4.1:

**Table 4.1: Classification Report - SVM**

| | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **0** | 0.99 | 0.99 | 0.99 | 273 |
| **1** | 0.99 | 0.99 | 0.95 | 323 |
| | | | | |
| **Accuracy** | | | 0.99 | 596 |
| **Macro Avg.** | 0.99 | 0.99 | 0.99 | 596 |
| **Weighted Avg.** | 0.99 | 0.99 | 0.99 | 596 |

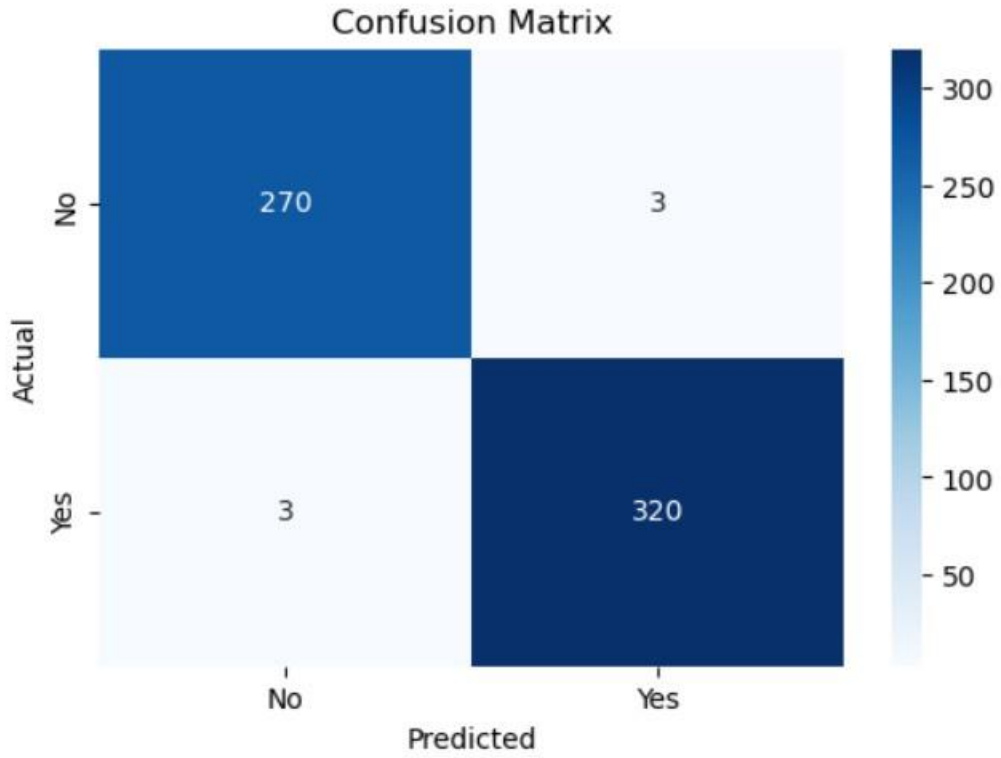


**Figure 4.1: Confusion Matrix - SVM**

### 4.1.2 Decision Tree

A decision tree is a popular machine learning algorithm used for both classification and regression tasks. It is a tree-like model that makes decisions based on the features of the input data. The Accuracy, Precision, Recall, F1-Score and Support of SVM algorithm are shows in the Table 4.2 and the Confusion Matrix is shows in the Figure: 4.2:

**Table 4.2: Classification Report – Decision Tree**

| | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **0** | 0.98 | 0.99 | 0.98 | 273 |
| **1** | 0.99 | 0.98 | 0.99 | 323 |
| | | | | |
| **Accuracy** | | | 0.98 | 596 |
| **Macro Avg.** | 0.98 | 0.99 | 0.98 | 596 |
| **Weighted Avg.** | 0.98 | 0.98 | 0.98 | 596 |

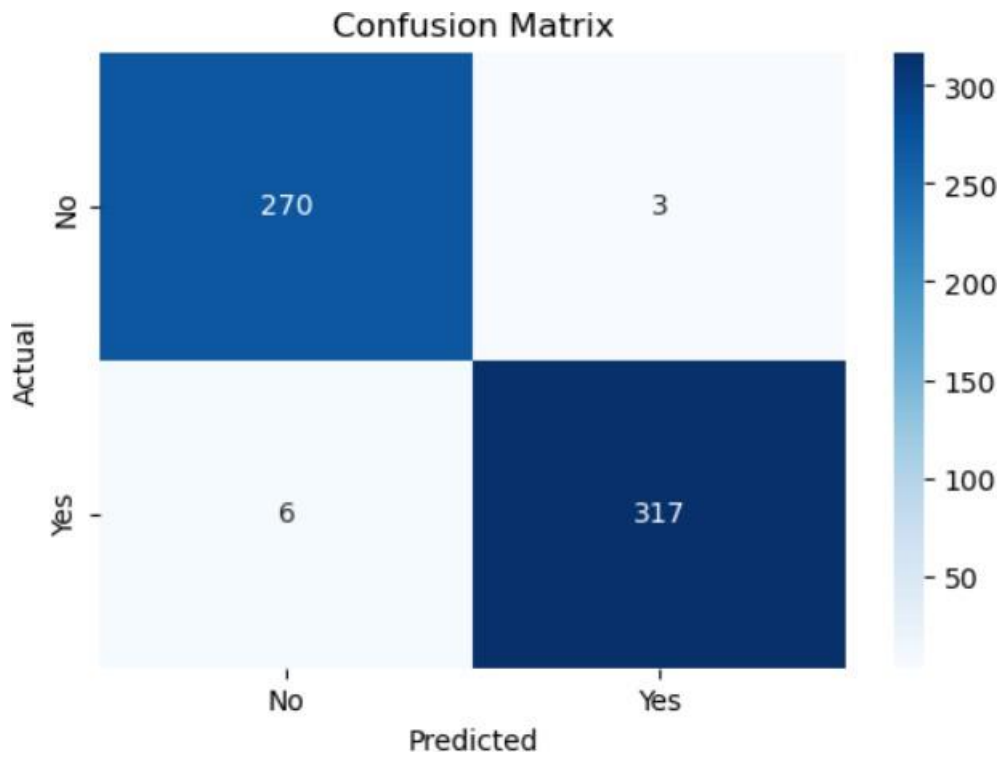


**Figure 4.2: Confusion matrix – Decision Tre**

### 4.1.3 Random Forest

A Random forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and control overfitting. It is widely used for both classification and regression tasks, leveraging the diversity of trees to enhance model performance. The Accuracy, Precision, Recall, F1-Score and Support of SVM algorithm are shows in the Table 4.3 and the Confusion Matrix is shows in the Figure: 4.3:

**Table 4.3: Classification Report – Random Forest**

| | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **0** | 0.81 | 0.95 | 0.87 | 273 |
| **1** | 0.95 | 0.81 | 0.88 | 323 |
| | | | | |
| **Accuracy** | | | 0.87 | 596 |
| **Macro Avg.** | 0.88 | 0.88 | 0.87 | 596 |
| **Weighted Avg.** | 0.86 | 0.87 | 0.87 | 596 |

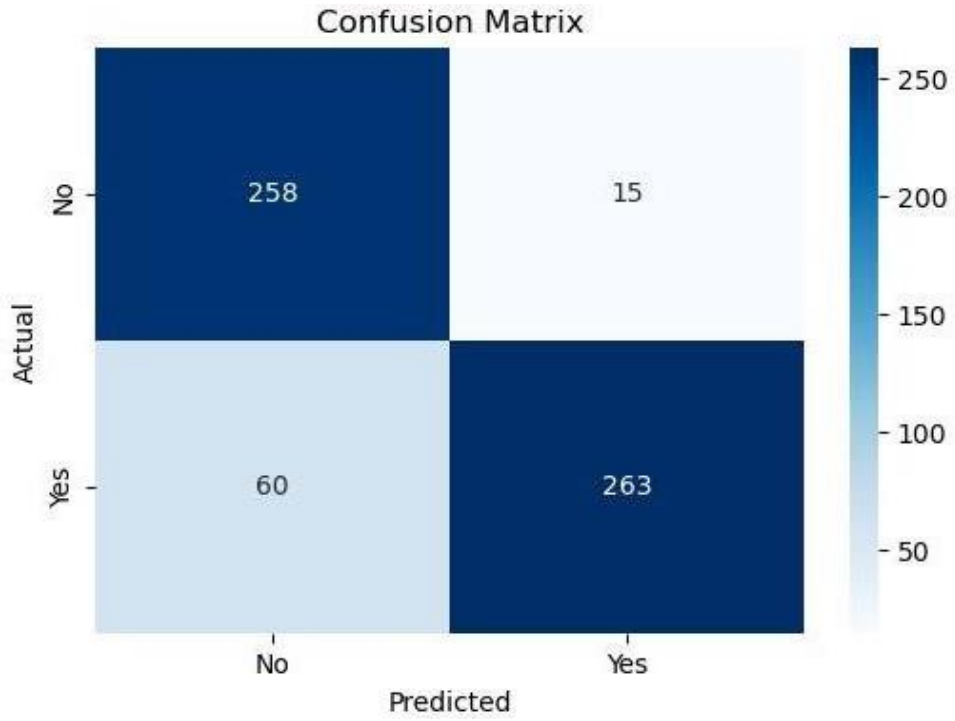


**Figure 4.3: Confusion matrix – Random forest**

### 4.1.4 Naïve Bayes

Naive Bayes is a family of probabilistic algorithms based on Bayes' Theorem,used primarily for classification tasks. Naive Bayes is a probabilistic machine learning algorithm based on Bayes' Theorem, with a strong (naive) assumption that all features are independent of each other given the class label. It is particularly effective for large datasets and is known for its simplicity and efficiency. The "naive" aspect of the algorithm comes from the assumption that the features are independent given the class label, which is often not the case in real-world data. Despite this assumption, Naive Bayes can perform surprisingly well in practice. Despite the simplicity, Naive Bayes can perform surprisingly well, especially with text data.The Accuracy, Precision, Recall, F1-Score and Support of SVM algorithm are shows in the Table 4.4 and the Confusion Matrix is shows in the Figure: 4.4

**Table 4.4: Classification Report – Naïve Bayes**

| | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **0** | 0.68 | 0.76 | 0.72 | 273 |
| **1** | 0.77 | 0.70 | 0.73 | 323 |
| | | | | |
| **Accuracy** | | | 0.72 | 596 |
| **Macro Avg.** | 0.73 | 0.73 | 0.73 | 596 |
| **Weighted Avg.** | 0.73 | 0.72 | 0.73 | 596 |

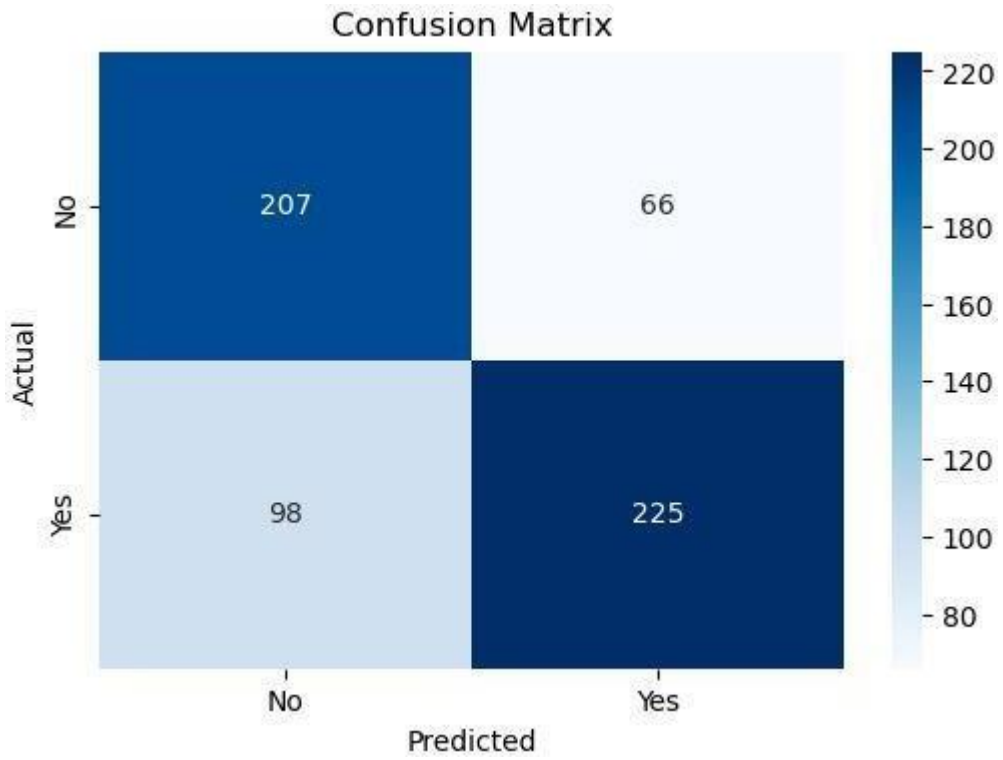


**Figure 4.4: Confusion matrix – naïve Bayes**

### 4.1.5 Logistic Regression

Logistic regression is a statistical method used for binary classification problems in machine learning. It models the relationship between a dependent binary variable and more independent variables by estimating probabilities using a logistic function. logistic regression is used for classification tasks rather than regression tasks. The Accuracy, Precision, Recall, F1-Score and Support of SVM algorithm are shows in the Table 4.5 and the Confusion Matrix is shows in the Figure: 4.5:

**Table 4.5: Classification Report – Logistic Regression**

| | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **0** | 0.96 | 0.96 | 0.96 | 273 |
| **1** | 0.97 | 0.96 | 0.96 | 323 |
| | | | | |
| **Accuracy** | | | 0.96 | 596 |
| **Macro Avg.** | 0.96 | 0.96 | 0.96 | 596 |
| **Weighted Avg.** | 0.96 | 0.96 | 0.96 | 596 |

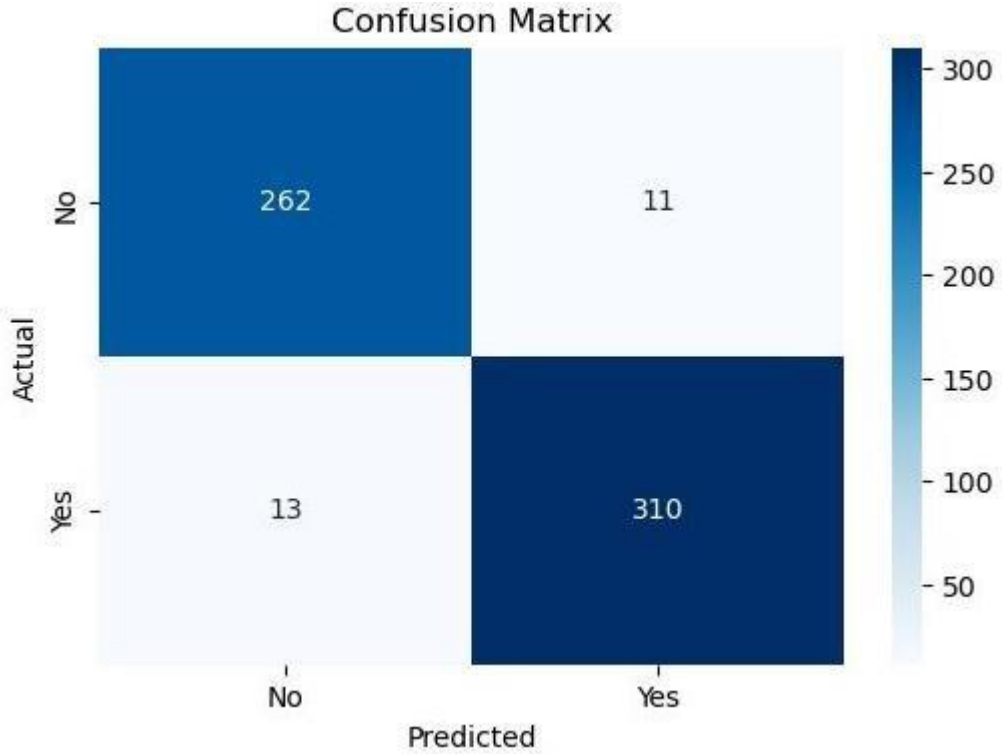


**Figure 4.5: Confusion Matrix – Logistic Regression**

## 4.2 ANALYSIS REPORT

Based on the evaluation metrics, the **Support Vector Machine (SVM)** algorithm demonstrated the **highest performance** with an **accuracy of 99%**, making it the most effective model for diagnosing Autism Spectrum Disorder (ASD) in this project. Its high score suggests that SVM effectively captures complex decision boundaries, particularly in high-dimensional data, and correctly classified almost all instances. We have taken five algorithms for the Diagnosis of Autism Spectrum Disorder. Machine learning Techniques. The Algorithm and its Accuracy are given in Table 4.6.

**Table 4.6 Algorithm and its Accuracy**

| | | |
|---|---|---|
| **SVM** | **-** | **0.99** |
| **Decision Tree** | **-** | **0.98** |
| **Random Forest** | **-** | **0.87** |
| **Naïve Bayes** | **-** | **0.72** |
| **Logistic Regression** | **-** | **096** |

**Support Vector Machine (SVM)** stands out as the most suitable algorithm for ASD diagnosis, followed closely by **Decision Tree and Logistic Regression**, depending on whether model interpretability or performance is prioritized.

# CHAPTER 5

# CONCLUSION AND FUTURE ENHANCEMENT

## 5.1 Conclusion

Previous research on Autism Spectrum Disorder (ASD) has predominantly focused on specific diagnostic tools, clinical assessments, or isolated machine learning approaches. While valuable, such studies often lacked a comprehensive integration of various algorithmic techniques and practical application through user-centric tools. This project addressed that gap by implementing a broad spectrum of machine learning algorithms—including SVM, Decision Tree, Random Forest, Naive Bayes, and Logistic Regression—for the early diagnosis of ASD using child screening datasets. The development of an interactive Streamlit web application further demonstrates the real-world applicability of this research, enabling caregivers and healthcare professionals to access predictions and model insights with ease. Ultimately, this project highlights the potential of data-driven solutions in supporting early detection and intervention strategies for ASD, contributing meaningfully to both academic research and clinical practice.

## 5.2 FUTURE ENHANCEMENT

In future work, we aim to explore more advanced and diverse machine learning and deep learning techniques to improve the accuracy and robustness of Autism Spectrum Disorder (ASD) prediction. While this study utilized binary classification (1 - ASD, 0 - non-ASD) based on existing screening features, future research can include multi-class classification to identify different severity levels or subtypes of ASD. Additionally, incorporating longitudinal data, behavioral patterns, and neuroimaging features may help in early developmental tracking and personalized diagnosis. Expanding the dataset to include more demographic diversity and real-time clinical inputs will further enhance the generalizability and reliability of the predictive system